

NPG Annealings

Phase 04 | Operator explanation and control logic

Release	Build	Reviewed	Executable
0.9.41	2026.09.04-r20	2026-09-04	Phase 04

Safety boundary: this guide describes the software sequence only. Follow the current laboratory SOP, confirm the physical chamber and instruments, and remain supervised. Software limits do not replace hardware interlocks or the emergency procedure.

Purpose

Complete the two-stage NPG annealing recipe and supervised cooldown after the Phase 03 electrical handoff.

Scope and ownership

The phase controls the external oven PID setpoint and the Keysight ramp-down while recording process values, stability windows, hold timers and final safe state. The oven controller is physically autonomous, so supervision remains required.

Operator sequence

1	Confirm the Phase 03 handoff, physical chamber condition and that the oven PID and Keysight connections are available.
2	Start Phase 04. The initial setpoint is 200 deg C and the default initial wait is 5 min.
3	The oven is driven to 350 deg C. The first 15 min hold begins only after the PV is inside +/-2 deg C for 30 continuous seconds.
4	The oven is then driven to 600 deg C. The second 40 min hold uses the same +/-2 deg C / 30 s stability qualification.
5	Hold timers pause when the PV is outside the configured stability band. Follow the live stage and remaining-time display.
6	After the annealing recipe, the oven PID is set to 0 deg C and the 10 min cooldown phase is recorded. The Keysight handoff current ramps down from 0.640 A in 0.005 A steps every 15 s after its configured delay.
7	Wait for the final output-off and cooldown checks, then allow the phase to save and close automatically.

Packaged defaults

Stage	Target	Stability	Hold
Initial wait	200 deg C	Recipe-controlled	5 min
First anneal	350 deg C	+/-2 deg C for 30 s	15 min
Second anneal	600 deg C	+/-2 deg C for 30 s	40 min
Cooldown	0 deg C	PID signal verified	10 min

Electrical parameter	Default
Phase 03 handoff current	0.640 A, output ON
Ramp-down step	0.005 A every 15 s
Output at zero	OFF

Stability and cooldown

A hold timer counts only while the oven process value remains inside the configured target band. A temporary excursion pauses progress rather than silently counting time outside the qualified window. The dashboard reports the active stage, target, stability state and known timed work remaining.

The external oven PID remains a separate physical controller. Setting its software setpoint to 0 deg C is not proof that the chamber is already cold. Remain present until the PID signal and physical condition confirm the expected cooldown.

Never leave the workstation solely because the automatic recipe has finished. Verify output OFF, PID setpoint 0 deg C, process value and the physical chamber state before ending supervision.

Finish and abort behavior

Normal completion	Abort / Safe Stop
Normal completion saves the final report and closes automatically after the cooldown window and output-off checks pass.	Abort requests the configured safe PID reset and completes the Keysight safe-zero/output-off path. Inspect the physical chamber before any further action.

Data written

A new folder is reserved below Data Samples/NPG Annealing Data using the pattern NPG Annealings <sample> data NN.

Important record	Use
Stage telemetry	Oven PV/setpoint, Keysight current/voltage and stage timing.
Stability records	Entry, exit and paused intervals for each annealing hold.
Final files	Final plots, summary and safe-state completion information.

Handoff and verification

When the phase closes, review the final summary and confirm that the launcher has released all configured serial ports. Preserve the run folder with the laboratory record; the allocator will create a new numbered folder for the next run.